

UNIVERSITY OF NEW HAMPSHIRE

**Monte Carlo Assessment of Two-Sided 8 MeV
Electron Irradiation of Cryogenic ND₃ for
Polarized Target Preparation**

Project Report

Samuel B. Takwirira

Nuclear Physics Group

University of New Hampshire

Jefferson Lab Target Group Collaboration

May 2026

Abstract

A TOPAS/Geant4 Monte Carlo model was used to assess whether a two-sided 8 MeV electron irradiation protocol can prepare packed cryogenic ND_3 for polarized-target use. The model includes the beam-spreading window, cryostat windows, liquid-helium bath, target-holder proxy, packed ND_3 target volume, front/back electron sources, dose scorers, fluence scorers, spectral diagnostics, and total energy-deposition checks. Because the target length is 6.5 cm, the calculation treats irradiation as a sequential front/back protocol rather than a one-sided exposure. The cold 1 μ A case delivers $4.97 \times 10^{15} \text{ e}^- \text{ cm}^{-2}$ per side, gives a mean two-sided dose of $2.63 \times 10^6 \text{ Gy}$, and deposits 0.064 W per active side. The warm 10 μ A case delivers $9.97 \times 10^{16} \text{ e}^- \text{ cm}^{-2}$ per side, gives a mean two-sided dose of $5.29 \times 10^7 \text{ Gy}$, and deposits 0.644 W per active side. The simulation supports the feasibility of a front/back irradiation geometry while leaving radical chemistry, ESR response, annealing behavior, and final DNP polarization to experimental benchmarks.

Keywords: polarized target, dynamic nuclear polarization, ND_3 , electron irradiation, TOPAS, Geant4, dose, fluence, cryogenic heat load.

1. Introduction

Irradiated ammonia targets require radiation-induced paramagnetic centers to support dynamic nuclear polarization (DNP) [3-6]. For ND_3 , useful target preparation depends not only on delivered charge, but also on where ionization dose is deposited, how uniformly the target volume is covered, and how much heat is deposited during cold irradiation. This report evaluates a proposed 8 MeV electron irradiation at Jefferson Lab UITF using Monte Carlo transport. The central question is whether the beam and geometry can deliver a proposal-scale radiation field throughout a 6.5 cm packed ND_3 target while maintaining a credible cold heat load.

2. Methods and Analysis Framework

The simulation was implemented in TOPAS/Geant4 [1,2]. The geometry contains the beam-spreading aluminum window, cryostat windows, liquid-helium bath, holder proxy, packed ND_3 target region, and downstream scoring/dump elements. The beam is an 8 MeV electron source rastered over a 14.3 mm radius field, corresponding to approximately 6.42 cm^2 interaction area. The packed target is modeled as a 6.5 cm long cylindrical volume with radius 1.42 cm and proposal-scale mass 250 mg [7].

The target length makes a one-sided exposure inadequate for the full material volume. The protocol is therefore modeled with an upstream front exposure and a downstream back exposure. The back-side dose is oriented into the same target coordinate system before summing,

$$D_{\text{FB}}(z) = D_{\text{front}}(z) + D_{\text{back,oriented}}(z).$$

Front and back irradiations are sequential, so heat load is interpreted per active side, not as a simultaneous two-sided load.

Scorers record depth dose, radial dose, R-Z dose maps, fluence through the target, entrance electron spectra, exit photon spectra, and total energy deposition. Per-primary Monte Carlo quantities are scaled by beam current and irradiation time. Dose rate is proportional to I/e , and deposited power is total target energy deposition per primary multiplied by the electron rate. The cold operating point is $1\ \mu\text{A}$ for 1.42 h per side; the warm operating point is $10\ \mu\text{A}$ for 2.85 h per side.

3. Results and Analysis

All graphical results are provided in the posted PowerPoint slides rather than embedded in this report.

3.1 Depth Dose and Two-Sided Coverage

The posted front/back depth-dose slide shows that the front-only dose is range-limited over the 6.5 cm target. The back-side exposure complements the front exposure after coordinate reorientation, and the summed front/back curve is the relevant target-volume dose field. For the cold case, the mean dose is $1.32 \times 10^6\ \text{Gy}$ per side and $2.63 \times 10^6\ \text{Gy}$ after the two-sided exposure. The R-Z and radial-uniformity slides show the spatial checks used to assess target-volume coverage rather than only the central beam axis.

3.2 Fluence, Spectra, and Heat Load

The electron-fluence slide connects Monte Carlo transport to the proposal fluence targets. The cold case delivers $4.97 \times 10^{15}\ \text{e}^- \text{cm}^{-2}$ per side and $9.94 \times 10^{15}\ \text{e}^- \text{cm}^{-2}$ for the two-sided exposure. The warm case delivers $9.97 \times 10^{16}\ \text{e}^- \text{cm}^{-2}$ per side and $1.99 \times 10^{17}\ \text{e}^- \text{cm}^{-2}$ for the two-sided exposure. The spectral slide verifies the target entrance electron spectrum and provides exit photon/bremsstrahlung context. The heat-load slide gives 0.064 W per active side for the cold case and 0.644 W per active side for the warm case.

3.3 Quantitative Summary

Table 1: Proposal-scaled fluence, dose, and heat load.

Mode	Delivered fluence per side	Two-sided fluence	Mean two-sided dose	Heat load per active side
Cold, $1\ \mu\text{A}$	4.97×10^{15}	9.94×10^{15}	$2.63 \times 10^6\ \text{Gy}$	0.64 W
Warm, $10\ \mu\text{A}$	9.97×10^{16}	1.99×10^{17}	$5.29 \times 10^7\ \text{Gy}$	6.44 W

Delivered fluence is within approximately one percent of the proposal target values for both operating points.

4. Discussion

The central result is that the target geometry should not be interpreted with a one-sided 8 MeV exposure. Electron range and energy loss require a sequential front/back protocol for proposal-relevant coverage of the packed ND_3 volume. The cold heat-load estimate is a promising feasibility indicator, but its practical meaning depends on the refrigerator, thermal coupling, holder design, and beam structure.

The model is a beam-transport and energy-deposition calculation. It predicts dose, fluence, spectra, spatial coverage, and deposited power; it does not predict radical species, radical survival, ESR spectra, annealing, or final DNP polarization. Those material-response quantities require experimental benchmarks, especially ESR characterization and NMR or direct polarization measurements after the proposed irradiation and thermal history.

A normalization limitation is that this Monte Carlo model used the nominal 250 mg batch mass distributed over the full holder volume, corresponding to an effective density of about 0.006 g cm^{-3} . The more relevant packed-pellet density is closer to 0.5 g cm^{-3} after applying the packing fraction. This difference explains why the present dose-rate estimate is likely too high: if too little mass is assigned to the scoring volume, the same deposited energy is converted into a larger Gy s^{-1} value. The same issue can affect heat-load normalization. A follow-up calculation will therefore rerun the model using a packed-pellet density and geometry details from the posted slides of Dr. Chris Keith, then compare the result directly with ESTAR-based estimates.

5. Conclusion

The Monte Carlo model supports continued development of a two-sided 8 MeV irradiation protocol for cryogenic packed ND_3 . The front-only dose field is range-limited, while the front/back protocol provides target-volume coverage consistent with the proposal fluence scale. The cold operating point gives $4.97 \times 10^{15} \text{ e}^- \text{ cm}^{-2}$ per side and an estimated 0.064 W heat load per active side. The next step is experimental validation of radical production, survival, thermal behavior, and final DNP performance.

Acknowledgments

This report was prepared by Samuel B. Takwirira for the University of New Hampshire Nuclear Physics Group and the Jefferson Lab Target Group collaboration. The author thanks Dr. Karl Slifer for guidance during critique sessions.

Data and Code Availability

The TOPAS inputs, analysis scripts, scorer outputs, and figure-generation materials are archived in the project repository: <https://github.com/SamuelBarachel/UITF02>. Numerical values in this report are tied to the archived production outputs used for this analysis.

References

- [1] S. Agostinelli et al., “GEANT4 - A simulation toolkit,” *Nuclear Instruments and Methods in Physics Research Section A*, vol. 506, no. 3, pp. 250-303, 2003. doi: 10.1016/S0168-9002(03)01368-8.
- [2] J. Perl, J. Shin, J. Schumann, B. Faddegon, and H. Paganetti, “TOPAS: An innovative proton Monte Carlo platform for research and clinical applications,” *Medical Physics*, vol. 39, no. 11, pp. 6818-6837, 2012. doi: 10.1118/1.4758060.
- [3] D. G. Crabb and W. Meyer, “Solid polarized targets for nuclear and particle physics experiments,” *Annual Review of Nuclear and Particle Science*, vol. 47, pp. 67-109, 1997. doi: 10.1146/annurev.nucl.47.1.67.
- [4] W. Meyer, “Ammonia as a polarized solid target material - a review,” *Nuclear Instruments and Methods in Physics Research Section A*, vol. 526, issues 1-2, pp. 12-21, 2004. doi: 10.1016/j.nima.2004.03.145.
- [5] W. Meyer, K. H. Althoff, W. Havenith, and W. Thiel, “Dynamic deuteron polarization in irradiated D-ammonia (ND_3) and its first use in a high energy photon beam,” *Nuclear Instruments and Methods in Physics Research Section A*, vol. 227, no. 1, pp. 35-44, 1984. doi: 10.1016/0168-9002(84)90098-6.
- [6] A. Conover, V. Y. S. Bandara, D. Keller, and F. Bateman, “Bulk irradiation of ammonia for polarized target experiments,” *Nuclear Instruments and Methods in Physics Research Section A*, vol. 1068, article 169717, 2024. doi: 10.1016/j.nima.2024.169717.
- [7] UNH-UITF BeamNetUS Proposal, internal project document.